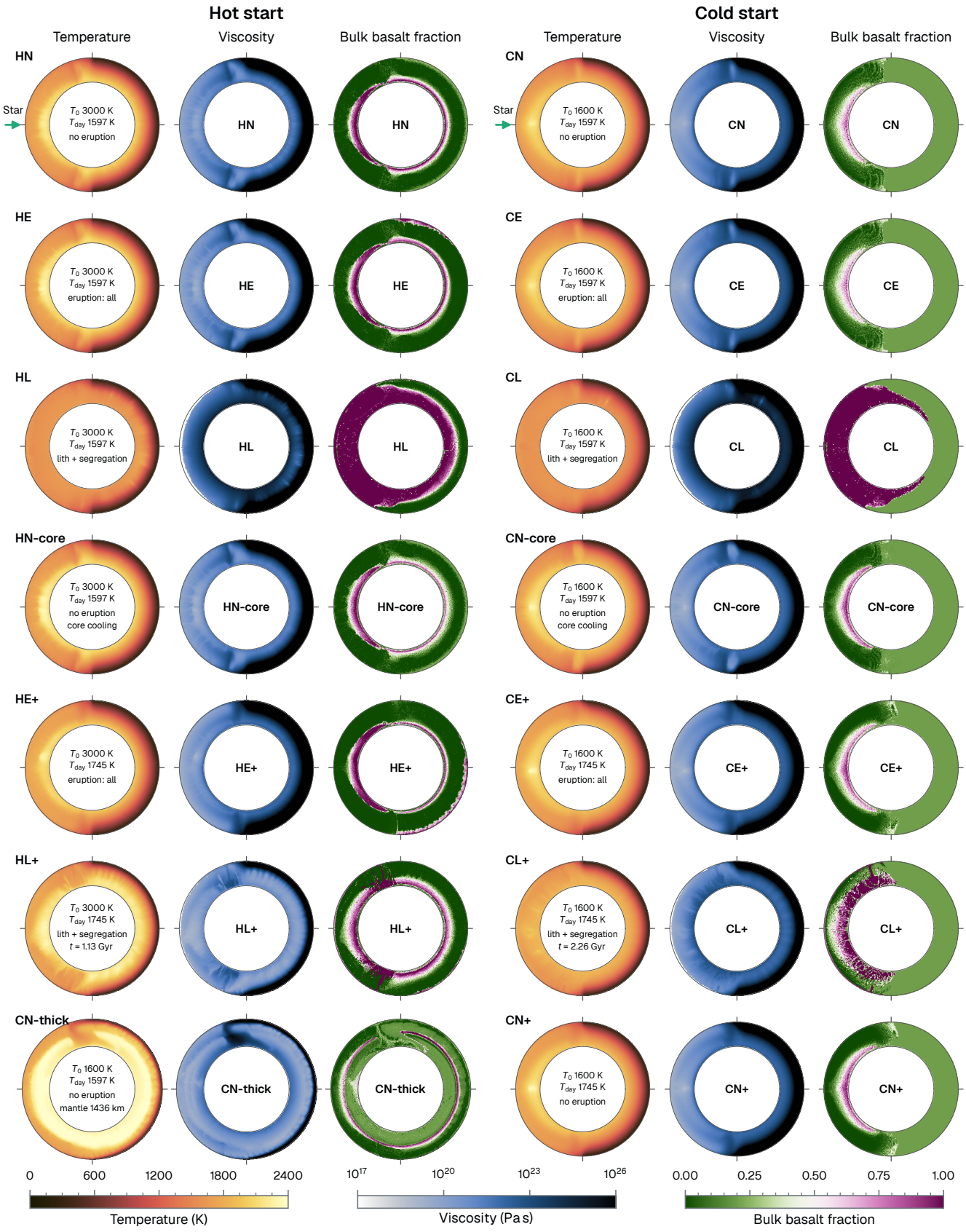

Hot start
Cold start
Temperature
Viscosity
Bulk basalt fraction
HN
Star
T_0 3000 K
T_{day} 1597 K
no eruption
HE
T_0 3000 K
T_{day} 1597 K
eruption: all
HL
T_0 3000 K
T_{day} 1597 K
lith + segregation
HN-core
T_0 3000 K
T_{day} 1597 K
no eruption
core cooling
HE+
T_0 3000 K
T_{day} 1745 K
eruption: all
HL+
T_0 3000 K
T_{day} 1745 K
lith + segregation
t = 1.13 Gyr
CN-thick
T_0 1600 K
T_{day} 1597 K
no eruption
mantle 1436 km
CN
T_0 1600 K
T_{day} 1597 K
no eruption
CE
T_0 1600 K
T_{day} 1597 K
eruption: all
CL
T_0 1600 K
T_{day} 1597 K
lith + segregation
CN-core
T_0 1600 K
T_{day} 1597 K
no eruption
core cooling
CE+
T_0 1600 K
T_{day} 1745 K
eruption: all
CL+
T_0 1600 K
T_{day} 1745 K
lith + segregation
t = 2.26 Gyr
CN+
T_0 1600 K
T_{day} 1745 K
no eruption
0
600
1200
1800
2400
Temperature (K)
10^{17}
10^{20}
10^{23}
10^{26}
Viscosity (Pa s)
0.00
0.25
0.50
0.75
1.00
Bulk basalt fraction